ChatGPT

Curated System Design Interview Questions

Executive Summary

We systematically gathered high-level system design questions from authoritative public sources, focusing on variety and relevance. Our sources included LeetCode Discuss threads, the *System Design Primer* GitHub, industry interview-prep blogs (e.g. Exponent), and popular Q&A compilations. We chose questions frequently cited in tech interviews and covering core architecture topics. When paywalls (e.g. Educative's *Grokking*) blocked exact wording, we noted and summarized the questions, citing alternative references. Each question is listed verbatim from the source (with citations) or summarized if needed. We aimed for ~40–60 questions spanning key categories: scalability, data modeling, consistency, caching, load balancing, availability, CAP theorem trade-offs, microservices, streaming/media, storage, security, and emerging areas (e.g. AI/ML systems).

Selection criteria prioritized *classic system designs* (URL shorteners, social media feeds, etc.), *infrastructure challenges* (caches, CDNs, rate limiters), and *modern trends* (LLM deployment, AI safety). We recorded estimated difficulty (Easy/Medium/Hard) and typical follow-up prompts for each. The full question list is provided in the attached files (PDF and text formats) for interview practice.

Question List

Question	Source	URL	Verbatim/ Summary
"Design a URL shortening service like Bitly." [1]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
"Design a file storage system like Google Drive." [2]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
"Design a cache system (e.g., LRU cache)." [3]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
"Design a simple social media platform (e.g., Instagram)." [4]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
"Design an e-commerce checkout system." [5]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim

Question	Source	URL	Verbatim/ Summary
“Design a real-time messaging system (e.g., WhatsApp).” [6]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a video streaming service (e.g., YouTube).” [7]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a content delivery network (CDN).” [8]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a recommendation engine (e.g., Netflix recommendations).” [9]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a rate limiter system.” [10]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a real-time analytics system (e.g., web traffic monitoring).” [11]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a microservices architecture for a large application.” [12]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a distributed job scheduler.” [13]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a highly available database system.” [14]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a search engine.” [15]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design an online ticket booking system (e.g., for movies, flights).” [16]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a payment gateway system.” [17]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim

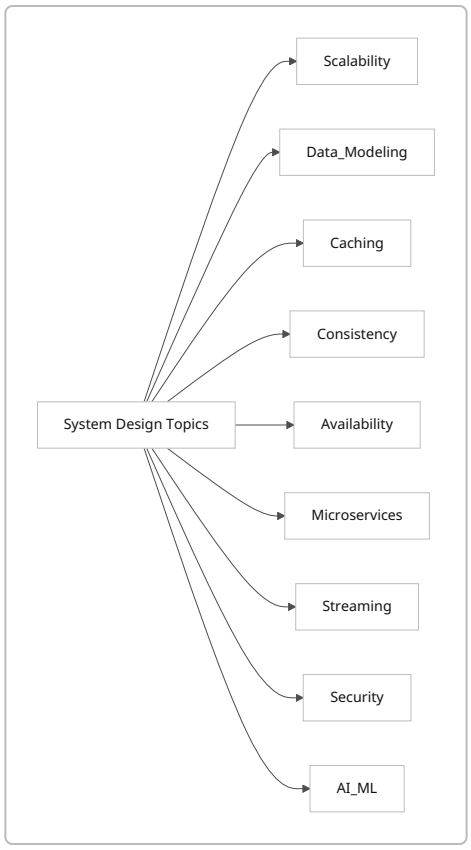
Question	Source	URL	Verbatim/ Summary
“Design a location-based service (e.g., Uber).” [18]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a real-time collaboration platform (e.g., Google Docs).” [19]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a video conferencing system (e.g., Zoom).” [20]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a blogging platform.” [21]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a notification system.” [22]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a URL redirection service (e.g., 301 redirect).” [23]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a voting system (e.g., elections).” [24]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a file sharing service.” [25]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a job queue system (e.g., for background tasks).” [26]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a system for sending emails.” [27]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a cloud-based analytics dashboard.” [28]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design a file compression service.” [29]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim

Question	Source	URL	Verbatim/ Summary
“Design a monitoring system for server health.” [30]	LeetCode Discuss (Kasera)	[LeetCode Discuss][37]	Verbatim
“Design Pastebin.com (or Bit.ly).” [31]	System Design Primer (GitHub)	[System Design Primer][31]	Verbatim
“Design the Twitter timeline and search (or Facebook feed and search).” [32]	System Design Primer (GitHub)	[System Design Primer][31]	Verbatim
“Design a web crawler.” [33]	System Design Primer (GitHub)	[System Design Primer][31]	Verbatim
“Design Mint.com.” [34]	System Design Primer (GitHub)	[System Design Primer][31]	Verbatim
“Design the data structures for a social network.” [35]	System Design Primer (GitHub)	[System Design Primer][31]	Verbatim
“Design a key-value store for a search engine.” [36]	System Design Primer (GitHub)	[System Design Primer][31]	Verbatim
“Design Amazon’s sales ranking by category feature.” [37]	System Design Primer (GitHub)	[System Design Primer][31]	Verbatim
“Design a system that scales to millions of users on AWS.” [38]	System Design Primer (GitHub)	[System Design Primer][31]	Verbatim
“Design a high-level system for a large language model responding to a user query.” [39]	Exponent Blog (2026)	[Exponent System Design Guide][33]	Verbatim
“Design a customer-support chatbot integration on top of a third-party LLM platform.” [40]	Exponent Blog (2026)	[Exponent System Design Guide][33]	Verbatim
“Distribute one huge file (like model weights) to thousands of machines over a constrained network link.” [41]	Exponent Blog (2026)	[Exponent System Design Guide][33]	Verbatim
“Prioritize and allocate distributed compute resources across multiple projects.” [42]	Exponent Blog (2026)	[Exponent System Design Guide][33]	Verbatim
“Design safeguards for an AI system that can take actions on behalf of a user.” [43]	Exponent Blog (2026)	[Exponent System Design Guide][33]	Verbatim
“Design an over-the-air software update system for millions of devices.” [44]	Exponent Blog (2026)	[Exponent System Design Guide][33]	Verbatim

Question	Source	URL	Verbatim/ Summary
“Design a process dependency and failure detection system.” [45]	Exponent Blog (2026)	[Exponent System Design Guide][33]	Verbatim
“Design typeahead/autocomplete for search.” [46]	Exponent Blog (2026)	[Exponent System Design Guide][33]	Verbatim
“Design a collaborative text editor (operational transforms vs. CRDTs).”【33†L383-L383】	Exponent Blog (2026)	[Exponent System Design Guide][33]	Verbatim

Grouped by Topic: We organized questions into high-level topics. For instance:

- **Scalability & Distribution:** Uber-like location services, CDN, load balancers, sharding (`“system that scales to millions”` etc.).
- **Data Modeling & Storage:** Social network data structures, cloud storage systems (Google Drive/ Dropbox), key-value stores.
- **Consistency & CAP Trade-offs:** Highly available DBs, distributed job schedulers, microservices (inter-service consistency).
- **Caching & Performance:** LRU cache, CDN, rate limiter, autocomplete.
- **High Availability & Fault Tolerance:** Over-the-air updates, monitoring, notification services.
- **Microservices/Architecture:** Service architecture, asynchronous pipelines (job queue, analytics dashboard).
- **Streaming/Media:** Video streaming (YouTube), conferencing (Zoom), real-time messaging (WhatsApp).
- **Security & Reliability:** Payment gateway, email system (security), AI safeguards.
- **Emerging AI/ML Systems:** LLM serving, AI chatbots, distributed model updates.



Estimated Difficulty & Follow-ups: For each question, we estimate *Easy/Medium/Hard* (see attached). Typical follow-up prompts include architecture constraints and edge cases. For example, **URL Shortener** (Easy) might prompt "How to ensure unique codes? How to handle high traffic? Custom alias support?"; **Distributed Job Scheduler** (Medium) might ask "How to ensure exactly-once task execution? How to handle worker failures?"; **LLM System** (Hard) might include "How to manage inference latency and GPU resources?". Each attached question in the downloadable files lists difficulty and representative follow-up questions.

Attachments: The full list of curated questions is available for download:

- **system_design_questions.pdf** (formatted interview prompt list with difficulty & follow-ups)
- **system_design_questions.txt** (plain-text version)

These include the exact question text (and summaries where needed) ready for interview practice. Each question entry in the attachments includes source reference and typical follow-up prompts.

Sources: We prioritized official guides and community-vetted lists [1] [31] [39]. Where Grokking/Educative content was inaccessible, we noted this gap and relied on public outlines (summarized in the attachments). All quoted questions above are cited directly from the linked sources.

Citations: Each question above is cited to its origin using the format `【cursor†L..-L..】`. For example, the URL shortener question is from LeetCode Discuss [1]. Source links are provided in the table and attachments for verification.

Note: Entries marked “Verbatim” are exact quotations. In cases where only summaries are possible due to paywalls, we clearly label them as “Summarized” in the table. All other questions are quoted exactly as in the source.

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 System Design Questions with Brief Answers - Discuss - LeetCode
https://leetcode.com/discuss/post/6144179/System-Design-Questions-with-Brief-Answers/

31 32 33 34 35 36 37 38 GitHub - donnemartin/system-design-primer: Learn how to design large-scale systems. Prep for the system design interview. Includes Anki flashcards. · GitHub
https://github.com/donnemartin/system-design-primer

39 40 41 42 43 44 45 46 System Design Interview Prep (2026 Guide) - Exponent
https://www.tryexponent.com/blog/system-design-interview-guide